

ESO208 COMPUTATIONAL METHODS FOR ENGINEERS

ASSIGNMENT 3

MAXIMUM MARKS: 30

DUE DATE: 29 AUGUST 2026

Q.1. [15 marks] Consider the following function

$$f(x) = e^{(x-1)^2} - 1$$

- (a) Plot the functions $f(x)$, $f'(x)$ and $f''(x)$. By looking at the plots, determine the multiplicity of the roots of $f(x)$ at $x = 1$. [3]
- (b) Plot the functions $u(x) = f(x)/f'(x)$ and $u'(x)$. By looking at the plots, determine the multiplicity of the roots of $u(x)$ at $x = 1$. [2]
- (c) Calculate the root of the function $f(x)$ using the Secant method and the modified Secant method using $x_{-1} = -0.5$ and $x_0 = 0$ as the initial guess. Compare the number of iterations required for convergence in the two methods. [10]

Q.2. [15 marks] (a) Write a code to calculate the root of a system of n linear equations with n unknowns using Gauss elimination with pivoting. [10]

(b) Use the code in (a) to calculate the root of the following system of equations. [5]

$$2x_1 - 6x_2 - x_3 = -38$$

$$-3x_1 - x_2 + 7x_3 = -34$$

$$-8x_1 + x_2 - 2x_3 = -20$$